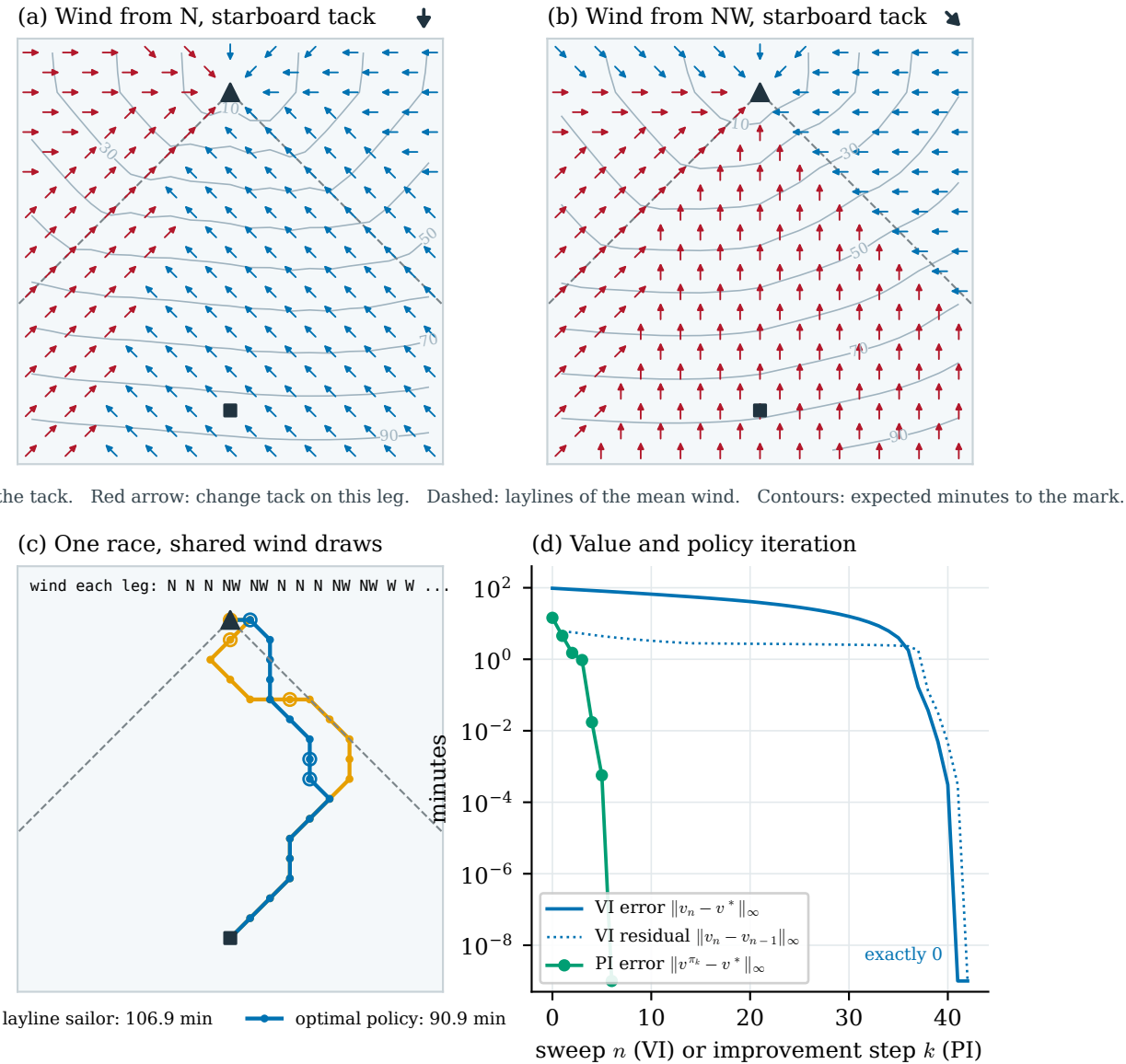

(a) Wind from N, starboard tack
(b) Wind from NW, starboard tack
10
30
50
70
90
Blue arrow: keep the tack. Red arrow: change tack on this leg. Dashed: laylines of the mean wind. Contours: expected minutes to the mark.
(c) One race, shared wind draws
wind each leg: N N N NW NW N N N NW NW W W ...
layline sailor: 106.9 min
optimal policy: 90.9 min
(d) Value and policy iteration
minutes
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
VI error $\|v_n - v^*\|_\infty$
VI residual $\|v_n - v_{n-1}\|_\infty$
PI error $\|v^{\pi_k} - v^*\|_\infty$
exactly 0
0
10
20
30
40
sweep n (VI) or improvement step k (PI)